MIL-C-22328A(Wep)
19 October 1964
Superseding
MIL-C-22328(Wep)
17 February 1960

MILITARY SPECIFICATION

CARTRIDGE, DELAY, MARK 31 MOD 0

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 This document governs the manufacture, assembly and preparation for delivery of the MARK 31 MOD 0 delay cartridge and the methods of inspection and tests upon which product acceptance will be based.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings, and publications, of the issue in effect on date of invitation for bids, form a part of this document to the extent specified herein.

SPECIFICATIONS

Military

MIL-M-21383	Manganese Delay Composition
MIL-P-22264	Powders, Ignition, Gasless
MIL-D-21625	Design and Evaluation of Cartridges for Cartridge Actuated Devices

STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

FSC 1375

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DRAWINGS

Bureau of Naval Weapons

LD 419691	Cartridge, Delay, MARK 31 MOD 0
LD 419692	Container, Inner, for Cartridge, Delay, MARK 31 Type
LD 419693	Container, Outer, for Cartridge, Delay, MARK 31 Type

PUBLICATIONS

SECNAV Instruction 4030.3B	Uniform Preservation, Packaging, Packing and Marking of Items of Military Supply

Bureau of Naval Weapons

OP 5	*Ammunition Ashore* (This publication is for the exclusive use of the Government inspector. It is not to be furnished to bidders or contractors.)
NAVORD Instruction 8010.24	Marking of Federal Stock Numbers and Nomenclature on Ammunition

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-90	Transportation, Interstate Commerce Commission, Explosives and Other Dangerous Articles

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Orders for the above publication should cite the latest edition and supplements thereto.)

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3. REQUIREMENTS

3.1 General.

3.1.1 The MARK 31 MOD 0 delay cartridge covered by this specification shall be manufactured in accordance with LD 419691 and all documents listed thereon.

3.2 Material. All materials used in the manufacture of the cartridges shall conform strictly with the specifications referred to on the respective BUWEPS drawings unless specific approval in writing covering a departure therefrom has been obtained from the Bureau of Naval Weapons prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal.

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, a preproduction sample of 84 MARK 31 MOD 0 delay cartridges shall be manufactured using the same methods and procedures proposed for the production lot. The sample shall be tested as specified in 4.3.2 and is for the purpose of determining that the contractor, prior to his starting production, is capable of producing items which comply with the technical requirements of the contract.

3.3 Primary components. For the purposes of this specification the manganese delay composition, MIL-M-21383, and Powder, Ignition, Gasless, MIL-P-22264, are defined as primary components.

3.4 Insulation resistance. Prior to assembly of the bridgewires, the insulation resistance shall not be less than 10 megohms at a relative humidity of less than 80 percent when measured with a 500 volt megohm meter. See 4.7.1.

3.5 Bridgewire. The bridgewire shall be resistance welded to the cartridge body and electrode.

3.6 Bridgewire assembly, continuity. The assembly shall be thoroughly cleaned to remove flux and contamination. Subsequent to cleaning, the bridgewire shall be flashed by application of an electric current to assure continuity of both legs of the bridgewire. The current shall be regulated to produce a visible red glow of the bridges. An EMF of approximately 1.5 volts AC or DC may be used with a suitable series resistor to adapt the source characteristics to the primer resistance.

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Flashing is a necessary manufacturing operation and shall be conducted on each bridgewire assembly. Assemblies which fail to glow in both legs of the bridge shall be screened from acceptable material in the process. The discarded assemblies may be reworked by replacement of the bridge.

3.7 Ceramic-to-metal seal. The ceramic-to-metal seal shall be in accordance with BUWEPS Dwg. 2085624.

3.8 Igniter composition and loading. The igniter composition shall be compounded and loaded in accordance with BUWEPS Dwgs. 2171815 and 1863093.

3.9 Gasless ignition powder. The gasless ignition powder A-1A shall be manufactured in accordance with MIL-P-22264 and loaded in accordance with BUWEPS Dwg. 2085630.

3.10 Manganese delay composition. The manganese delay composition shall be manufactured in accordance with MIL-M-21383 and loaded in accordance with BUWEPS Dwg. 2085630.

3.11 Primer (Relay). The primer (relay) shall be manufactured in accordance with BUWEPS Dwg. 2171816 and loaded in accordance with BUWEPS Dwg. 2085630.

3.12 Propellant charge.

3.12.1 Quantity. All propellant charges shall be as specified on Bureau of Naval Weapons Dwg. 1863093. Charge shall be determined by weight rather than by volume measurement.

3.12.2 Smokeless powder. Do not subject the smokeless powder to unnecessary handling or exposure to the atmosphere. Inspect for signs of decomposition as outlined in paragraphs 1308-1318 and 1322 of OP 5. The handling and loading of smokeless powder exposed to the atmosphere shall be performed only under conditions that do not permit the relative humidity to exceed 75 percent.

3.13 Cartridge inspection.

3.13.1 Visual inspection. The complete cartridge shall be free of the visible defects noted in 4.7.4.

3.13.2 Bridgewire resistance. The resistance of the bridges in each completely assembled cartridge manufactured, when measured electrically as specified in 4.7.3 shall be not less than 0.12 ohm or more than 0.18 ohm.

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3.14 Functional and ballistic requirements. Cartridges from each production lot, selected in accordance with 4.7.5, shall meet the requirements listed below when subjected to the test prescribed in 4.7.5.

3.14.1 Chamber pressure. The upper and lower limits of maximum chamber pressure shall be U = 21,000 and L = 10,500 psi respectively.

3.14.2 Delay time. Ignition delay, when the cartridge is actuated at 5 amperes per circuit input current, shall be not less than 0.50 seconds or more than 2.00 seconds.

3.14.3 Misfire. There shall be no misfires, see 4.7.5.3.

3.15 Workmanship. Cartridges shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention should be directed to dimensions, finishes and coatings.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling. Unless otherwise specified, and when applicable, the sampling plans and procedures used in the determination of the acceptability of products submitted by a supplier shall be in accordance with the provisions of MIL-STD-105. Unless otherwise specified in the contract, order, or requisition, all samples for test purposes shall be provided at the expense of the contractor.

4.3 Lots.

4.3.1 Inspection lots. Inspection lot definition, formation, and size shall be in accordance with MIL-STD-105.

4.3.2 Preproduction sample. Before entering into quantity production, an acceptable preproduction sample of 84 MARK 31 MOD 0 delay cartridges shall be prepared. A preproduction sample acceptable for functional and environmental testing shall be defined as a sample which has met the requirements of paragraphs 3.4 through 3.13.2 of this specification. The contractor shall deliver the cartridges to

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the activity designated by the contracting agency for test. Preproduction sample acceptance shall consist of the following tests as prescribed in Table III of MIL-D-21625: 40 ft drop (6), 15 g shock (12), vibration (12), temperature and humidity cycling (18), -65°F (12), +70°F (12), +160°F (12). The cartridges shall be fired in the test set of 4.7.5.1. No cartridge firing shall produce results outside the limits set forth in 3.14.1, 3.14.2 and 3.14.3. Any quantity production prior to notification by the contracting agency of preproduction sample acceptability shall be at the contractor's risk. The preproduction cartridges will not be applied as a part of the quantity specified for delivery by the contract.

4.3.3 Production lot. The suggested production lot size is 501 minimum and 1200 maximum cartridges. Cartridges required for all test purposes are selected from the production lot and will not be applied as a part of the quantity specified for delivery by the contract. Only primary components from a single lot shall be used in a production lot of cartridges. One primary component production lot may be used in more than one cartridge production lot. Cartridges selected for the ballistic and functional test shall be delivered to the activity designated by the contracting agency.

4.4 Gages. The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings.

4.5 Production and acceptance inspection. Production and acceptance inspection shall consist of the following:

Item	Nature of Test	Requirement	Test
Insulation	Resistance	3.4	4.7.1
Bridgewire assembly	Continuity	3.6	
Ceramic-to-metal seal	Condition	3.7	4.7.2
Propellant charge	Quantity	3.12.1	
Smokeless powder	Condition	3.12.2	
Cartridge	Visual inspection	3.13.1	4.7.4
Cartridge	Bridgewire resistance	3.13.2	4.7.3
Cartridge	Functional & ballistic	3.14	4.7.5

4.6 Packing, packaging and marking. The inspector shall ascertain that the packing and packaging of the cartridges and their markings conform to Section 5 of this specification.

4.7 Tests.

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4.7.1 Insulation resistance. Prior to assembly of the bridgewires, the insulation resistance shall be checked with a 500 volt megohm meter. Units with a resistance of less than 10 megohms shall be rejected.

4.7.2 Ceramic-to-metal seal. The ceramic-to-metal seal for each cartridge shall be tested in accordance with BUWEPS Dwg. 2085624 prior to bridgewiring.

4.7.3 Bridgewire resistance. Conformance with the bridge circuit resistance of 3.13.2 shall be ascertained by means of a test circuit which limits the bridge current to 50 milliamperes, maximum. A safety chamber with an interlock switch shall be used to protect the operator during the application of the current. A suitable bridge which is known to be accurate within 0.005 ohm in the range of resistance specified shall be used to determine whether or not the requirement is being met. Units which conform to the requirements of 3.13.2 shall be acceptable.

4.7.4 Visual inspection. Examine each cartridge produced under a contract and reject for: dents, deep scratches, split or cracked lip in the crimped area, perforated closure discs, sharp edges, burrs, incomplete threads and any other defects which may conceivably prevent entry into the firing chamber or affect stability or resistance to moisture.

4.7.5 Functional and ballistic tests.

4.7.5.1 The number of cartridges selected from each lot for the ballistic tests shall be in accordance with the requirements of Table I of MIL-STD-105, Inspection Level II. The cartridges shall be conditioned for 24 hours at +70°F ± 5°F and fired in a test chamber constructed in accordance with BUWEPS Dwg. 2022792 of LD 419691 or equivalent. Test firing current shall be 5 amperes per circuit. Pressure shall be recorded using a Wiancko Model P-1008 (20,000 psig) transducer or equivalent.

4.7.5.2 Criteria of acceptability. The acceptance number shall be zero and the rejection number shall be one using the limits prescribed in 3.14.1 for chamber pressure and 3.14.2 for delay time.

4.7.5.3 Misfire. The failure of any cartridge to fire shall result in rejection of the lot represented, unless such failure is plainly attributable to faulty test equipment or procedure.

4.8 Retests. There shall be no retests.

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4.9 Test failure. If test failure is attributable to an assignable cause, excluding the test cartridges, original test results should be discarded and the test reconducted with a new sample of cartridges.

5. PREPARATION FOR DELIVERY

5.1 Packaging. In addition to the following, Level A and Level C packaging shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.1.1 Level A. Level A packaging shall be used for packaging of all production lots for service use. For a definition of Level A packaging see 6.4.1.

5.1.1.1 Inner container. One cartridge shall be packaged in accordance with LD 419692.

5.1.2 Level C. Minimum packaging for those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be Level C. For a definition of Level C packaging see 6.4.1.1.

5.2 Packing. In addition to the following, Level A and Level C packing shall be in accordance with the Code of Federal Regulations 49 CFR 71-90.

5.2.1 Level A. Level A packing shall be used for packing of all production lots for service use. For a definition of Level A packing see 6.4.2.

5.2.1.1 Outer container. Ten inner containers as described in 5.1.1.1 shall be packed in accordance with LD 419693.

5.2 Level C. Minimum packing for those cartridges which are intended for expenditure in preproduction sample testing and for production lot acceptance tests shall be Level C. For a definition of Level C packing see 6.4.2.1.

5.3 Marking. Cartridge marking shall be as specified on Bureau of Naval Weapons Drawing 1863093 and NAVORD Instruction 8010.24. In addition, packaging and packing markings shall be in accordance with the explosive section of the Code of Federal Regulations 49 CFR 71-90, MIL-STD-129 and NAVORD Instruction 8010.24.

6. NOTES

6.1 Intended use. The MARK 31 MOD 0 delay cartridge is used to actuate a refueling hose guillotine which severs a refueling hose in an emergency.

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6.2 Ordering information. Procurement documents should specify the following:

a. title, number, and date of this specification;

b. whether preproduction inspection is required and if so specify the test activity;

c. production lot size and test activity (see 4.3.3);

d. items of data required for each preproduction and production lot (see 6.3).

6.3 Data. For the information of Contractors and Contracting Officers, any of the data specified in (a) subparagraphs below, (b) applicable documents listed in Section 2 of this specification or (c) referenced lower-tier documents, need not be prepared for the Government and shall not be furnished to the Government unless specified in the contract or order. The data to be furnished shall be listed on DD Form 1423 (Contractor Data Requirements List), which shall be attached to and made a part of the contract or order. NAVWEPS Form 4200/15 (Drawings, Lists, and Specifications Required) shall be attached where applicable.

6.3.1 Test Report. Results of ballistic tests shall be forwarded to the Bureau of Naval Weapons and the procuring activity.

6.3.2 Data Cards. The contract or order should state the format, quantity and destination of data cards.

6.4 Definitions. The definitions of Level A and C packaging and packing given below were extracted from SECNAV Instruction 4030.3B.

6.4.1 Level A packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment, handling, indeterminate storage and world-wide redistribution.

6.4.1.1 Level C packaging. The degree of preservation and packaging which will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity, for immediate use or controlled humidity storage. This level may conform to supplier's commercial practice when such meets the requirements of this level.

6.4.2 Level A packing. The degree of packing which will afford adequate protection during shipment, handling, indeterminate storage and world-wide redistribution.

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6.4.2.1 Level C packing. The degree of packing which will afford protection against damage during direct domestic shipment from the supply source to the first receiving activity for immediate use. This level in general will conform to applicable carrier rules and regulations and may be the supplier's commercial practice when such meets the requirements of this level.

6.5 General safety precautions.

6.5.1 The loading, assembly, and handling of explosives, ammunition, ammunition components, fuzes, detonators, primers, tracers, boosters, pyrotechnics, etc., and subassemblies thereof, involve hazardous operations and therefore require suitable explosive precautions. Use of this specification shall not be construed as to relieve the contractor or manufacturer of responsibility for the safety of his operations. Listed below are the minimum provisions which a contractor or manufacturer (loader) of the above items should observe in order to fulfill his responsibility for safety. At Bureau of Naval Weapons, Navy Department, and other Government plants, these provisions are mandatory. Such other warnings and precautions, pertinent to the operational effectiveness or safety during use or loading of the specified item, are included in the detail provision of the specification.

6.5.2 All loading operations should be conducted in a neat and orderly manner.

6.5.3 Safe equipment and methods should be utilized for transporting and handling explosives and loaded parts. Where required, remote control barricaded handling equipment shall be used for explosives operations, such as mixing, pouring, weighing, charging, sifting, drying, pressing, casting, crimping, etc.

6.5.4 Personnel handling detonators, primers, delay elements, lead-ins, boosters, and related parts which affect functioning should insofar as practicable, avoid using bare fingers or improper equipment in order to prevent damage, corrosion, or deterioration from perspiration or other contaminating deposits.

6.5.5 The exposure of explosive materials and related parts should be so controlled as to minimize the absorption of moisture from the atmosphere or other sources during loading and handling operations.

6.5.6 All explosives and completely or partially loaded items should be stored in suitable storage magazines located in accordance with the American Table of Distances (ATD) or other applicable safety standards; and, while in process, in safety lockers and chests if in

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loading rooms, or in adequate ready or service magazines located in accordance with intraplant distances when outside of loading rooms. For Navy explosives loading plants, the provisions of the Armed Services Explosive Safety Board covering quantity-distance regulations for explosives, shall apply.

6.5.7 Proper care should be exercised at all times to protect personnel from accidents, fires, or explosions, and to limit damage to equipment and loading areas. In this connection, the following precautionary measures should be observed:

6.5.7.1 Employ properly proportioned and properly located protective barricades, screens or shields at all required points.

6.5.7.2 Keep only minimum limited quantities of explosives and completed or partially loaded parts present at each stage of operation.

6.5.7.3 Keep explosives and explosive parts in approved covered receptacles with covers in place when material is not being taken out of or put into the receptacles. Where necessary, receptacles should be conductive to ground electrostatic charges.

6.5.7.4 Protect operations from electrostatic charges by effectively grounding all machinery, equipment, and fixtures; and, when necessary, employ suitable grounded conductive coverings for floors, work benches and tables, and workers' conductive shoes. Workers' clothing of a type to minimize the accumulation of static charges should be employed. Fabrics such as silk and nylon, which promote static generation should be avoided. Additional grounding devices such as grounded bracelets for workers should be employed where operations are conducted with items which are unusually sensitive to initiation by static electricity. Such items include initiating explosives, tracer mixtures, and low-energy type electric primers, detonators and squibs. The latter types of items should have the free ends of lead wires bared and twisted together, and be packed in relatively small groups wrapped in bare non-insulated aluminum foil or other uncoated metal foil. During assembly and processing operations such sensitive electric items should be short circuited by clips or other devices until installed with safety shunt in the final device. Additional precautions for these items should include mechanical shielding to contain or deflect fragments and blast, also electrical shielding of these items from induced electric currents generated by sources such as lightning, static, radiations from communications apparatus, radar, or high frequency heating apparatus, etc. Where necessary for safety, humidity (see para. 6.5.5) of work rooms should be increased, as required to lessen electrostatic effects without excessive moisture absorption.

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6.5.7.5 Protect all explosive operations from effects of electric current originating from equipment such as soldering irons, heaters, switches, wiring, motors, lights, test instruments, etc., by suitable insulation, grounding, separation or shielding. Such electric sources may initiate explosives by heat, sparks, arcs, or by completing an electric circuit through an electric primer, detonator, or squib. Circuits may be inadvertently completed, for example, from a defective electric soldering iron through a grounded contact. All electric types of primers, detonators or squibs provided with wire leads should have the free ends of the wires bared and twisted together to short circuit each unit, except when in process of assembly into a finished item. Where practicable, removable short circuiting clips or other devices should be employed during manufacturing operations involving electric primers, detonators or squibs.

6.5.7.6 Enforce, where necessary, the wearing of suitable safety footwear, gloves, goggles, respirators, and impregnated garments to protect personnel against burns, poisoning and associated industrial hazards.

6.5.7.7 Allow no fires or exposed electrical or other sparking equipment, and little or no flammable material to be present in loading, handling and storage spaces. Enforce proper "Match" and "No Smoking" rules where necessary.

6.5.7.8 Enforce good housekeeping and maintain effective policing, inspection and supervisory methods throughout the loading area and surroundings. Employ effective cleaning methods periodically to minimize the accumulation of explosives or explosive dust and other contamination upon, and assure its removal from floors, walls, ceilings, ledges, tables, benches, piping, and equipment or the items loaded; also, clean up any spilled material immediately.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-005808H(Weps) COVERALL, FLYING, ANTI-EXPOSURE, QUICK DONNING, TYPE QD-1

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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